

Food Labels, Cognition, and Healthy Choices: Evidence from Experimental and Observational data

We examine evidence from randomised controlled trials and population-level surveys to understand the relationship between label design and eating behaviours.



by **Constanza Avalos**

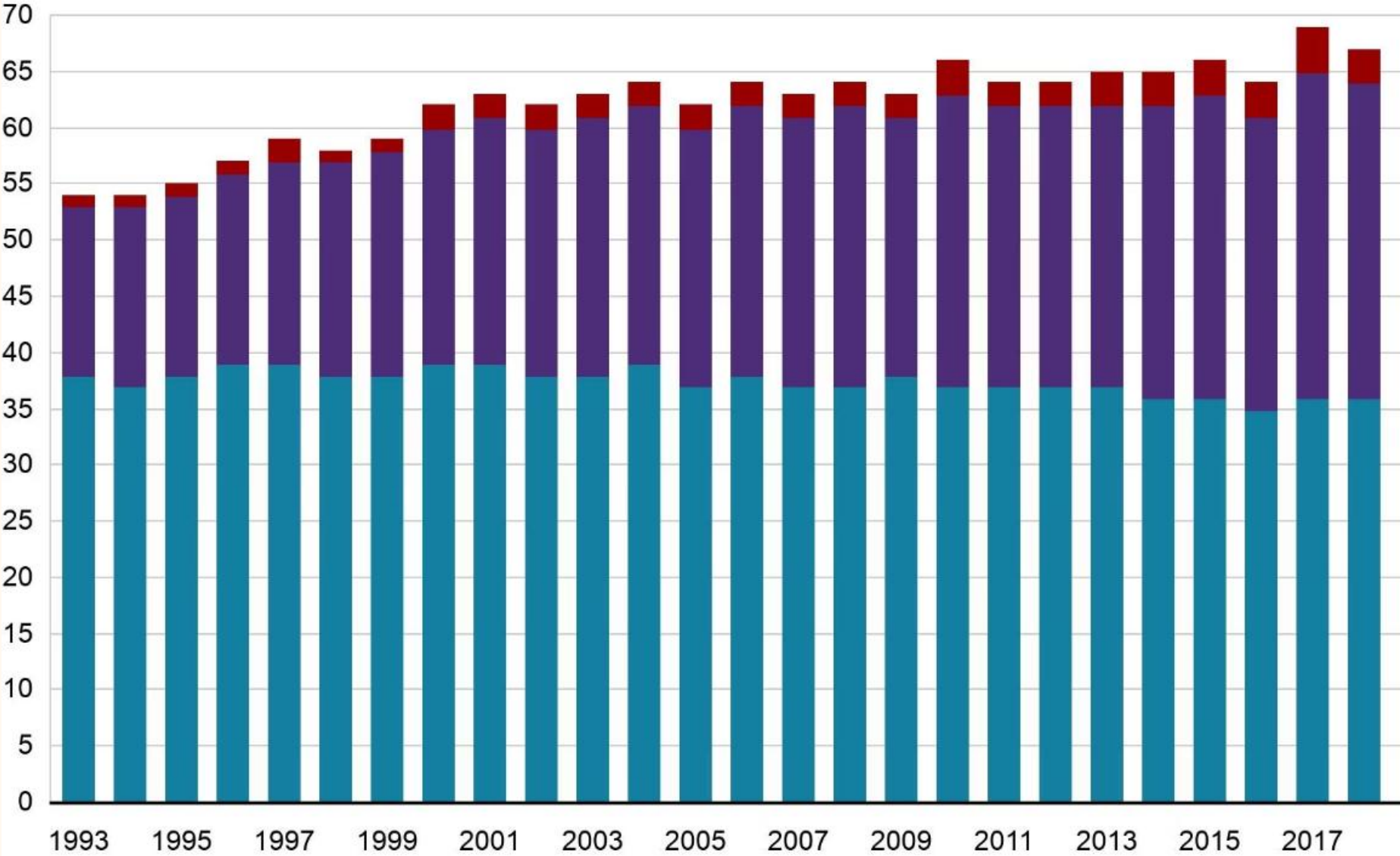
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How has obesity changed over time?

% of population over 16 in England

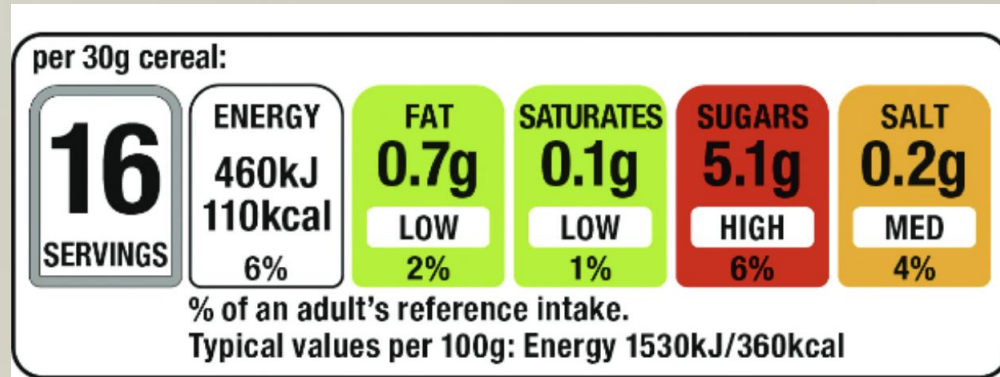
Overweight Obese Morbidly obese



Source: NHS Digital, Health Survey for England

BBC

Multiple Traffic Light (MTL)



Key Tool

Food labels guide consumers toward healthier dietary choices. They translate complex nutrition data into accessible information.



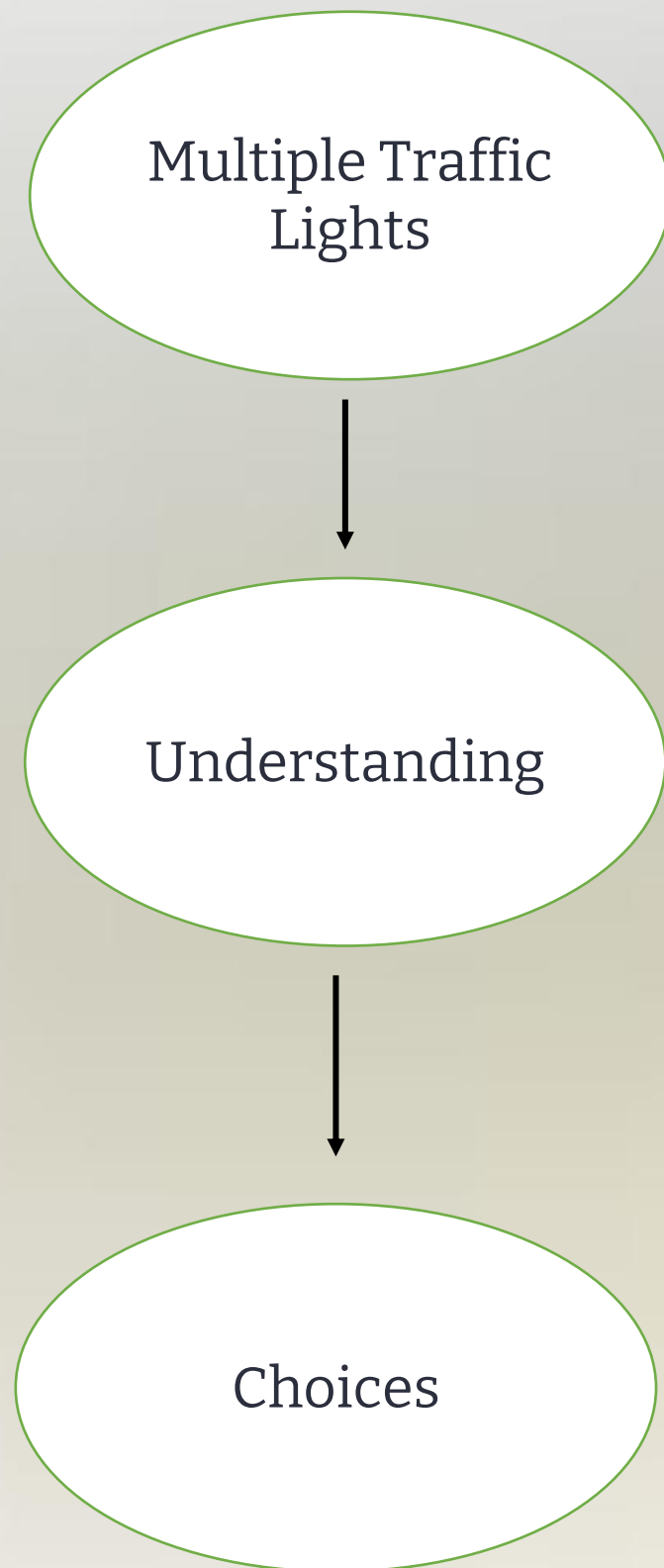
Growing Relevance

UK processed food consumption has increased by 50% between 2012-2018. This trend highlights the importance of effective labelling.



Cognitive Factors

This policy mitigates the negative consequences associated with high processed food intake by improving individual's understanding of nutritional information.



PhD research



GAP

There remains a gap in the literature regarding which existing label graphic designs are most interpretable for consumers.



Aims

I investigate whether the graphic design of MTL labels is legible and understandable for consumers, and whether this understanding translates into healthier food choices.



Methodology

I examine evidence from a randomised controlled trial and population-level survey using the **causal inference framework**.

Paper 1: Effects of Food Label Granularity and Working Memory

Randomised Controlled Trial

Compared detailed (fine-grained) versus coarse food labels on consumption decisions. Sample 498 participants.

Modelling

Multilevel linear and logistic regressions.

Key Finding

Detailed labels (8 levels) led to smaller calorie count than coarse labels.

Cognitive Moderation

Working memory capacity influenced how effectively people used label information.

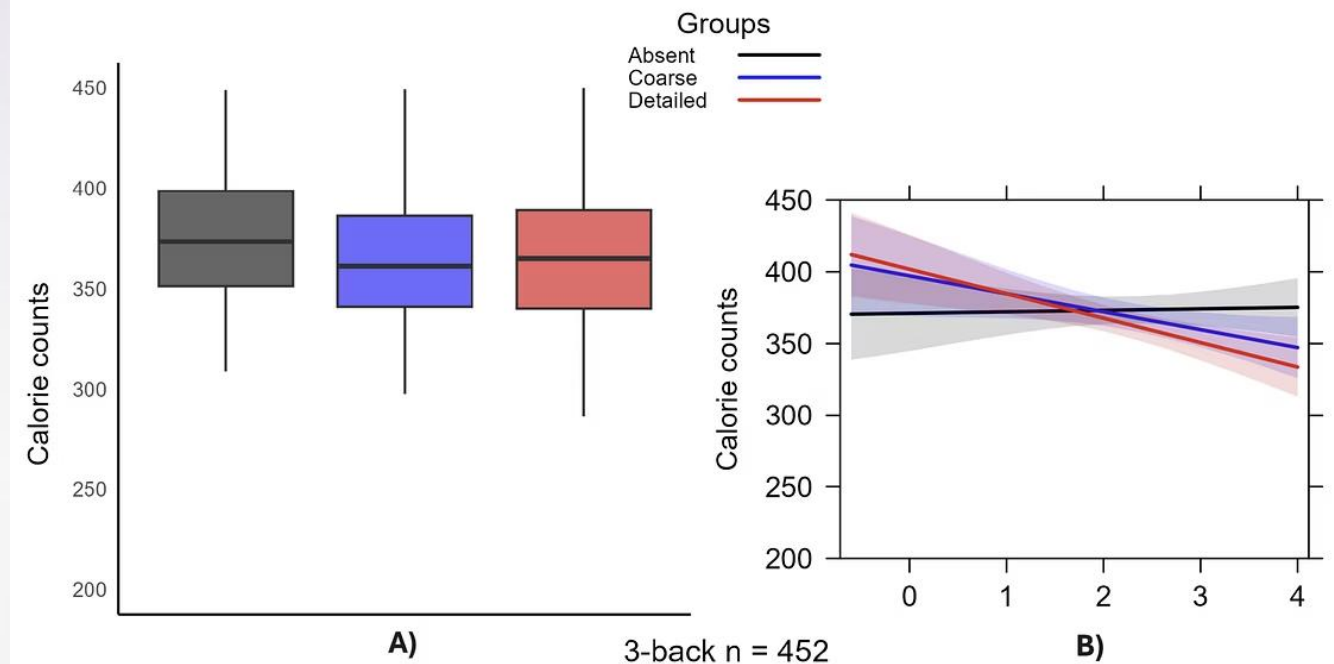
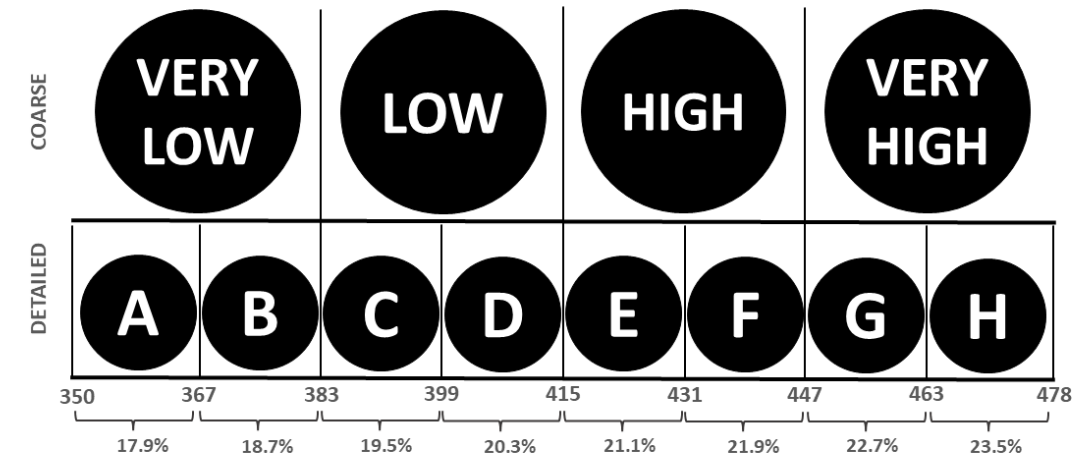


Fig. 5 A) Effects of FOP labelling on calorie counts by experimental conditions, including 3-back test performance levels as an interaction term. B) Plot showing the interaction effects between d' as 3-back performance and different labelling conditions on caloric count.

Paper 2: Multiple Traffic Light Label Readability

Table 5 Ordinal logistic regression results: associations of perceived MTL readability and ready-to-eat meal consumption frequency (adjusted for sociodemographic controls and years)

Variable	Pre-package sandwich			Pre-cooked meat		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
(Intercept):Weekly-Monthly	-0.57 (0.06)***	-1.14 (0.21)	-1.12 (0.27)***	-1.43 (0.09)***	-1.03 (0.26)***	-1.54 (0.30)***
(Intercept):Monthly-Never	0.95 (0.09)***			-0.40 (0.07)***		
Readability:Weekly-Monthly	0.15 (0.02)***	0.03 (0.02)	0.20 (0.07)**	-0.02 (0.03)	0.01 (0.03)	-0.07 (0.08)
Readability:Monthly-Never	0.21 (0.03)***	0.04 (0.03)	0.17(0.09)*	0.01 (0.02)	0.02 (0.02)	-0.01 (0.07)
Years:Weekly-Monthly	—	—	-0.05 (0.01)**	—		0.08 (0.07)
Years:Monthly-Never	—	—	0.04 (0.02)	—		0.21 (0.06)***
Readability:Years:Weekly-Monthly	—	—	-0.04 (0.07)*	—		0.02 (0.02)
Readability:Monthly-Never	—	—	-0.06 (0.02)**	—		0.01 (0.02)
Sociodemographics	—	Yes	Yes	—	Yes	Yes
N	8854	8854	8854	8854	8854	8854

Note: Significance: * < 0.05, ** < 0.01, *** < 0.001. Models 1-6 use svy_vglm with cumulative logit. Models account for survey design. N = number of participants.

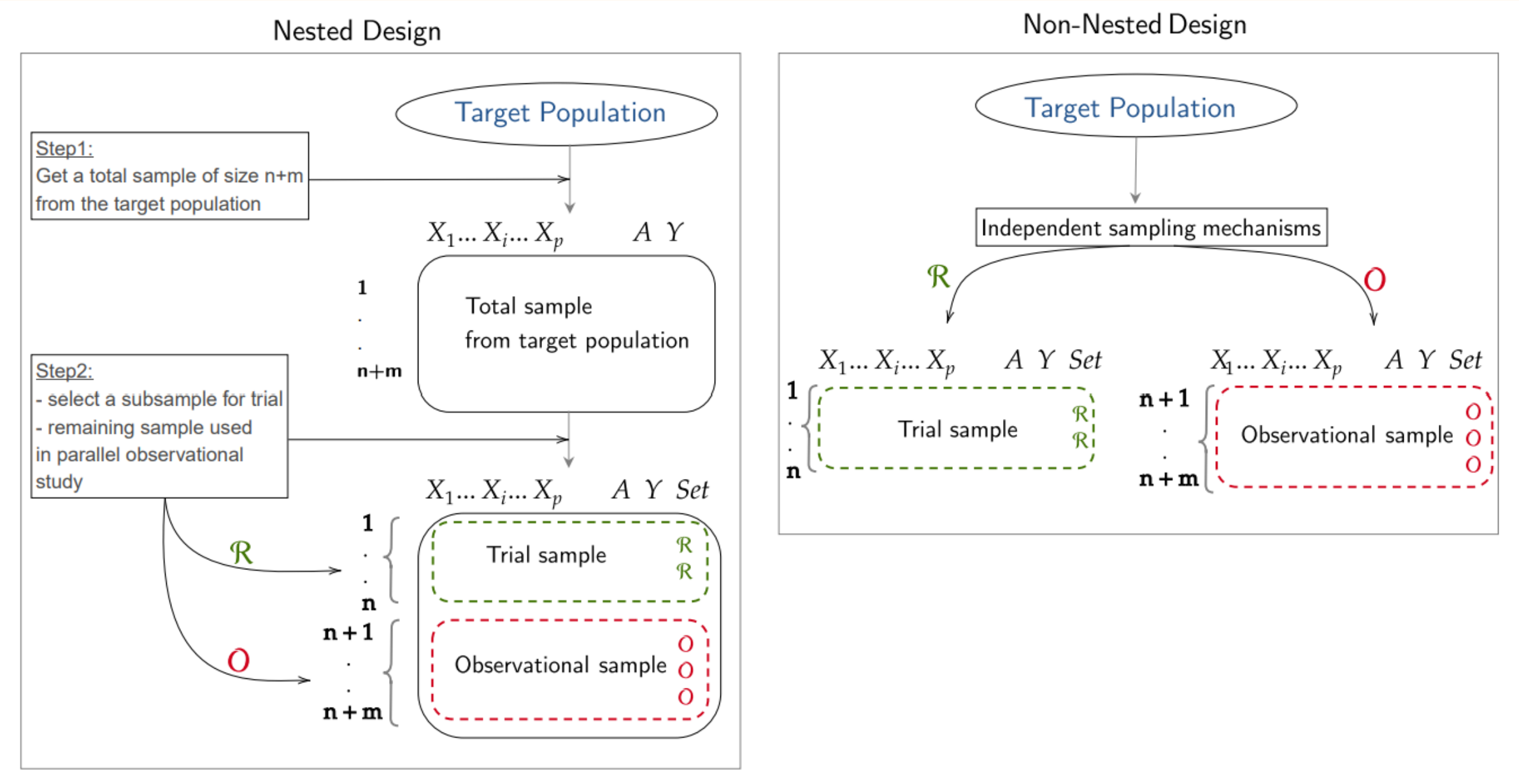
I investigate whether of label readability are associated with how often individuals consume ready-to-eat meals.

- We used responses from the Food and You Survey 2012 to 2018.
- Models: Ordinal logistic regressions.

Reduced readability scores correlated with higher ready-meal consumption frequency.

- Visual design clarity emerged as a powerful behavioral lever.
- Socio-demographic disparities in label comprehension and associations varies across survey waves

Paper 3: Transporting Food-Label Effects to UK Population



1 Key Estimate

Estimate average UK adult calorie savings from implementing trial label nationwide.

2 Methodology

Apply transportability methods to UK National Diet and Nutrition Survey (NDNS) 2009-2018.

3 Comparison

Transportability methods: inverse-odds-of-sampling weighting, outcome-model averaging, and doubly-robust.

Conclusions

Multiple Traffic Light labels convey nutritional information in an understandable way that leads to healthier food choices:

- Granular labels reduced calorie counts. While they didn't change choices to lower-calorie options, labelling increased selection of moderate-calorie cereals when considering the interaction with the most demanding n-back level.
- This variation represents a decrease of 3.88% in the coarse group (which resembles MTL) and 5.24% in the detailed group, where **4-5% represents approximately 100--110 calories** of a typical 2,000 calorie daily diet. This finding partially aligns with similar studies in the field.
- Perceived readability has a modestly significant association with dietary choices.
- Individuals who found the labels harder to read were about 3% more likely to eat pre-packaged sandwiches frequently each day, compared to those who ate them monthly. However, these associations vary across different groups and tend to diminish over time.

Limitations

RCT external validity. However, even a modest real-world impact would be a worthwhile public health intervention.

- 50% reduction in processed food consumption could prevent approximately 22,000 cardiovascular-related deaths in the UK by 2030

Expand Research

Thank you for your attention. Questions welcome.